

Full length article

MGTN-DSI: A multi-sensor graph transfer network considering dual structural information for fault diagnosis under varying working conditions[☆]Jianjie Liu^{a,b}, Xianfeng Yuan^{a,b},^{*} Xilin Yang^a, Weijie Zhu^{a,b}, Yansong Zhang^{a,b}, Tianyi Ye^{a,b}, Xinxin Yao^{a,b}, Fengyu Zhou^c^a School of Mechanical, Electrical & Information Engineering, Shandong University, Weihai 264209, China^b Shandong Key Laboratory of Intelligent Electronic Packaging Testing and Application, Shandong University, Weihai 264209, China^c School of Control Science and Engineering, Shandong University, Jinan 250100, China

ARTICLE INFO

Keywords:

Fault diagnosis

Multi-sensor

Unsupervised domain adaptation

Graph convolutional network

Varying working conditions

ABSTRACT

In recent years, mechanical systems have increasingly integrated multiple sensors to monitor equipment status more effectively. However, extracting domain-invariant features from multi-sensor data and adapting the diagnostic model to significant variations in operating conditions remain challenging tasks. To address these issues, a novel multi-sensor graph transfer network considering dual structural information (MGTN-DSI) is designed for fault diagnosis under varying working conditions. Firstly, we develop an advanced multi-sensor feature extraction mechanism. Specifically, on the one hand, a multi-sensor collaborative fusion layer is proposed to uncover the intrinsic connections among different sensor data. On the other hand, a relationship graph between data points of the fused high-level features is constructed using a graph convolutional network with constrained filters. Secondly, a joint alignment method using a virtual discriminator is proposed to simultaneously align both subdomain and global distributions. The extensive experiments conducted on a public fault diagnosis dataset and a practical fault diagnosis test platform indicate that the proposed MGTN-DSI has higher accuracy and better generalization ability than other state-of-the-art comparison methods.

1. Introduction

In recent years, with the rapid development of intelligent fault diagnosis technologies [1], deep learning (DL) has emerged as a powerful tool. It has been extensively applied in fault diagnosis across various fields [2,3], leveraging its robust feature extraction capabilities [4–6].

DL technology has been proven highly effective in fault diagnosis, driving research on various intelligent models and deep hybrid architectures [7–9], which combine multiple network types [10,11]. These methods exhibit significant potential in improving fault diagnosis accuracy and adapting to complex operating conditions, thus promoting the widespread application of intelligent diagnostic technologies across different industrial domains [12,13]. However, the assumption of identical distribution between training and testing sets, which underlies most intelligent fault diagnosis (FD) methods [14–16], is often challenging to achieve in practice, as variations in operating conditions, such as rotating speed and load, can significantly degrade diagnostic

performance [17,18]. Furthermore, labeling the health state of samples extracted from signals in these conditions is labor-intensive and time-consuming, making traditional intelligent FD methods unsuitable for real industrial environments.

Unsupervised domain adaptation (UDA) is a technique in deep learning, which has been applied to the case where the target domain (TD) lacks available labels under varying working conditions. It aims to improve the performance of the target domain by utilizing similar distributions across different domains and domain-invariant features from the labeled source domain (SD). For example, Wang et al. [19] proposed a TL model and utilized source samples to augment the absent classes in the TD, which effectively alleviates the domain bias due to carrying working conditions. Wan et al. [20] proposed a multilevel domain adaptation network employing attention mechanisms and hierarchical decoding, performing well in cross-machine bearing fault transfer diagnosis tasks. Similarly, Yu et al. [21] took the ResNet as the

[☆] This work was supported in part by the National Natural Science Foundation of China (Grant No. 61803227, 61973184), Shandong Provincial Natural Science Foundation, China (Grant No. ZR2023MF005), China postdoctoral science foundation (Grant No. 2023M743530), Young Scholars Program of Shandong University, Weihai (Grant No. 20820211010).

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<https://doi.org/10.1016/j.aei.2025.103119>

Received 18 July 2024; Received in revised form 14 December 2024; Accepted 6 January 2025

Available online 15 January 2025

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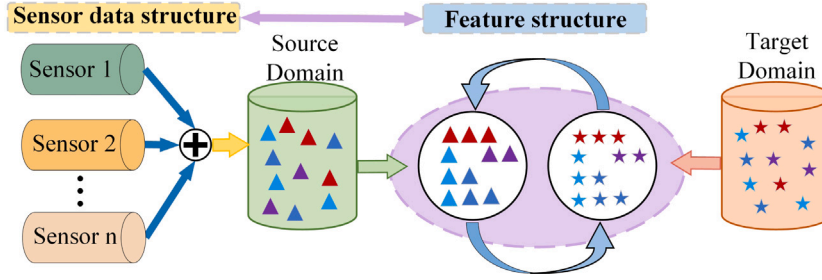


Fig. 1. Dual structure information in multi-sensor data.

extractor and transformed original signals to wavelet packets for better fault feature extraction, and further mitigated domain distribution discrepancies by multicore MMD. Besides the aforementioned metric-based techniques, adversarial-based UDA approaches belong to another mainstream of transfer FD methods. For instance, Xiao et al. [22] proposed a method of FD that employs the identification errors of source samples in the discriminator as weights in corresponding loss items for fine-grained feature alignment. Wu et al. [23] utilized Gaussian functions to guide distribution alignment and achieved cross-domain fault diagnosis tasks through an adversarial training mechanism. Embedding the Wasserstein distance in the domain discriminator, [24,25] effectively reduced conditional distribution divergence caused by changes in operating conditions.

In recent UDA-based fault diagnosis research, there has been a growing focus on uncovering latent spatial structure information within fault signals. Such structural information often reveals the intrinsic correlations between fault samples, providing a more comprehensive representation of domain-invariant features. Graph neural networks (GNNs) have gained significant attention [26] as a means to deeply explore the spatial structural information among fault data. For instance, Li et al. [27] proposed a graph transfer learning diagnosis framework, which constructs instance graphs and enhance domain-unique feature extraction. Yang et al. [28] considered cross-domain relationships by constructing domain graphs to extract the structure information in source and target data, achieving satisfactory cross-domain diagnosis results. Liang et al. [29] introduced a semi-supervised subdomain adaptation layer to align the distribution of the same fault category between two domains, enhancing the diagnosis accuracy in target samples.

Despite the considerable results achieved by the research mentioned above, there are still two key challenges: firstly, existing GNN-based domain adaptation methods primarily focus on single-sensor data, which restricts them to using only single structural information. However, with the rise of the Internet of Things, multi-sensor data is becoming increasingly prevalent in practical scenarios, offering a richer source of domain-invariant information. Meanwhile, as shown in Fig. 1, there exist topological relationships among multi-sensors at both the data level and the feature level. This introduces new challenges: accurately constructing structural relationships between different sensors and effectively fusing the high-level features from multiple sensors. Secondly, achieving a balance between domain alignment and feature discriminability poses a key challenge in UDA, particularly in multi-sensor scenarios where different sensors yield diverse representations of the same fault. This variability exacerbates the complexity of high-dimensional features in multi-sensor data, complicating both model training and domain alignment. Consequently, developing a method that enhances cross-domain consistency while maintaining category discriminability is crucial.

To address the aforementioned challenges, this paper proposes a novel domain adversarial graph neural network for multi-sensor fault diagnosis tasks under varying working conditions, namely, multi-sensor graph transfer network considering dual structural information (MGTM-DSI). Specifically, with the help of an advanced fusion strategy, the

graph convolutional network with a restricted filter-based extractor is able to extract the sensor relationship and the feature correlation simultaneously from multi-sensor signals. Moreover, a norm-based Wasserstein distance is embedded in the fault classifier, achieving adversarial training without discriminator and aligning conditional distribution divergence under varying working conditions. The main innovations and contributions are summarized as follows:

The main innovations and contributions are summarized as follows:

1. We propose a novel graph transfer network that utilizes multi-sensor data to extract domain-invariant features. This design incorporates a two-stage feature extraction process that achieves collaborative fusion across multiple sensors data, and the fused features are then transformed into relational graph representations, enhancing the learning of domain-invariant features. Additionally, a GNN with constrained filters is designed to enhance deep modeling of the sensor data space structure.
2. Considering the variability in feature representations of the same fault across different sensors, we present a joint domain alignment method based on adversarial and metric approaches. This method introduces a novel adversarial strategy, implemented through a virtual discriminator constructed by a classifier, thereby ensuring distinguishability between different domains while significantly enhancing the capability for feature alignment.
3. Comprehensive cross-domain diagnostic tests and ablation studies are conducted on both a widely used open access bearing fault diagnosis dataset and a practical fault diagnosis platform. Experimental results reveal that the proposed MGMN-DSI achieves superior performance in multi-sensor fault diagnosis under varying working conditions, outperforming the state-of-the-art comparison methods.

The structure of this paper is organized as follows: Section 2 introduces the theoretical framework. Subsequently, Section 3 elaborates on the proposed method. Section 4 discusses the experimental results and their analysis. Finally, the conclusion is drawn in Section 5.

2. Theoretical background

2.1. Problem definition

This study addresses fault diagnosis under varying working conditions, where the SD denoted by $D_s = (x_{11}^s, y_{11}^s), \dots, (x_{m n_s}^s, y_{m n_s}^s)$ has n_s labeled samples and m sensors, and the TD denoted by $D_t = \{x_{11}^t, \dots, x_{m n_t}^t\}$ has n_t samples that are not labeled in the TD. Additionally, the faulty samples are denoted as $x_i \in \mathcal{X}$, and the corresponding fault label is represented as $y_i \in \mathcal{Y}$. Meanwhile, $y_s = y_t = \mathcal{C}$, which indicates that the two domains share the same label space. The objective of the proposed MGMN-DSI is to learn the domain-invariant features from the SD, which is enabled to achieve satisfactory diagnostic results in TD tasks.

2.2. Typical domain adaptation methods

Maximum mean discrepancy [30] and Domain Adversarial Neural Networks [31] are the typical metric-based method and the adversarial-based method of unsupervised domain adaptation.

MMD is a nonparametric metric, frequently utilized to measure the consistency of the expectations of two distributions in the reproducing kernel Hilbert space (RKHS). MMD in RKHS can be defined as:

$$\mathcal{L}_{MMD} = \left\| \mathbb{E}_{P(x^s)}(\phi(x^s)) - \mathbb{E}_{P(x^t)}(\phi(x^t)) \right\|_2^2 \quad (1)$$

where $\phi(\cdot)$ represents the transformation from the original data space to RKHS. $P(x^s)$ and $P(x^t)$ are the marginal distributions of the source and target domains, respectively.

DANN took inspiration from generative adversarial networks, employing an adversarial framework between the feature extractor and a domain discriminator to capture transferable features between domains. The objective is to learn domain-invariant features that effectively deceive the domain discriminator. Through the optimization process, the feature extractor focuses on acquiring domain-invariant features while the discriminator learns to distinguish between the domains to which different features belong. With this adversarial learning model, the feature extractor learns cross-domain invariant representations of both domains. The formulation for DANN's objective function is given by:

$$\mathcal{L}(\theta_f, \theta_c, \theta_d) = \mathcal{L}_c(\theta_f, \theta_c) - \lambda_{DANN} \mathcal{L}_{DANN}(\theta_f, \theta_d) \quad (2)$$

where θ_f , θ_c , and θ_d represents the parameters of the feature extractor, the fault classifier, and the domain discriminator, respectively. $\mathcal{L}_c$ is the prediction loss using cross-entropy, and λ_{DANN} is the coefficient to balance the adversarial loss $\mathcal{L}_{DANN}$.

2.3. Chebyshev interpolation

Suppose a filter function $l(\hat{\lambda})$ is continuous in the interval $[-1, 1]$ and its value at a finite number of points $\hat{\lambda}_k$ is known. It is able to approximate this function using polynomial interpolation, i.e., $l(\hat{\lambda}_k) = P_K(\hat{\lambda}_k)$, where P_K is a K -order polynomial. It is a common practice to sample $K + 1$ points uniformly from the interval $[-1, 1]$, namely, $\hat{\lambda}_0 < \hat{\lambda}_1 < \dots < \hat{\lambda}_K$, to construct the interpolating polynomial $P_K(\hat{\lambda})$. Intuitively, the more sampling points offer chances for the more accurate approximation of the function. However, according to the Runge Phenomenon [32], the increasing of K does not always enhance the performance.

Different from the general interpolation above for uniform sampling, Chebyshev interpolation [33] uses Chebyshev nodes as interpolation points, achieving better approximation and faster convergence. Chebyshev nodes are essentially the zeros of the $(K + 1)$ -order Chebyshev polynomial. Given a filter function $l(\hat{\lambda})$ that is continuous, then the Chebyshev node can be calculated as $x_j = \cos\left(\frac{j+1/2}{K+1}\pi\right)$ for T_{K+1} , where $j = 0, \dots, K$, and $l(x_j)$ denotes the function value of node x_j . The Chebyshev interpolation of $l(\hat{\lambda})$ is defined as:

$$P_K(\hat{\lambda}) = \sum_{k=0}^K c_k' T_k(\hat{\lambda}), c_k = \frac{2}{K+1} \sum_{j=0}^K l(x_j) T_k(x_j) \quad (3)$$

where c_k denotes the interpolated Chebyshev polynomial coefficients.

3. Proposed method

3.1. Overall framework

Fig. 2 illustrates the detailed diagnosis process of the MGTN-DSI. During the model training phase, multi-sensor signals are first input to the weight-shared extractor, and fault features can be obtained from the convolutional neural network for SD and TD, respectively. With the help of the sensor fusion layer, the sensor relationships are constructed, which can be reflected by the fused compact features.

Table 1

Network parameter setting.

Block	Layers	Parameters
MFL	Conv1	In_channel=16, kernel=15, stride=1, BN
	Conv2	16x[32, kernel=3, stride=1], BN
	Maxpool	3 × 1 kernel, stride=2, ReLU
	Conv3	32x[64, kernel=3, stride=1], BN
	Conv4	64x[128, kernel=3, stride=1], BN
	Adaptive MaxPool	4, ReLU
	Linear	128 × 4, 256, ReLU
RFGNN	Fusion	256
	GConv1	256 × 512, K=[1,2,3], BN, ReLU
	GConv2	512 × 256, K=[2,1], BN, ReLU
Classifier/ Discriminator	Linear	256x Fault Categories

To further mine the feature relationships, the feature embeddings are then employed as node representations. Subsequently, they are input into the proposed GNN architecture with the restricted filter, further mining the feature relationship information from the node features with sensor relationship to obtain dual structure information. Moreover, the classifier is optimized to minimize the norm-based Wasserstein metric of fault features between SD and TD, achieving adversarial training with the extractor and extracting domain-invariant features. Finally, the classifier can accurately diagnosis the fault categories in the target domain. The extractor and classifier of MGTN-DSI are described as follows, and the complete algorithm flow is shown in Algorithm 1.

Algorithm 1 MGTN-DSI-based fault diagnosis method

- 1: **Input:** Dataset X with length l ; Number of sensors M ; Weights α and β ; Number of iterations k .
- 2: **Output:** Diagnosis results Y of the datasets X
- 3: Dividing dataset X into labeled source domain set $\{(\{x_{im}^s\}_{m=1}^M, \{y_{im}^s\}_{m=1}^M)\}_{i=1}^{n_s}$ and unlabeled target domain set $\{(\{x_{im}^t\}_{m=1}^M)\}_{i=1}^{n_t}$.
- 4: Constructing and Training MGTN-DSI model:
- 5: **for** $j = 1, 2, \dots, k$ **do**
- 6: Obtain the compact representation h of the multiple signals is obtained by the MFL
- 7: Construct association graphs based on KNN, to obtain adjacency matrices A_K
- 8: $[Q_i^s, Q_i^t] \leftarrow \text{RFGNN}([h_i^s, h_i^t], A_K)$
- 9: Set loss function:
- 10: $L(Q_i^s, Q_i^t, y_i^s) \leftarrow L_C(Q_i^s, y_i^s) + \alpha L_{DA}(Q_i^s, Q_i^t) + \beta L_{MMD}(Q_i^s, Q_i^t)$
- 11: Update the network parameters using the Adam optimizer.
- 12: **end for**
- 13: Testing the trained MGTN-DSI using the test data

3.2. Feature extractor considering dual structure information

The extractor comprises two components: the multi-sensor fusion layer and the restricted filters based graph convolutional network. The former constructs the sensor relationship, while the latter establishes the feature relationship for fault signals. The network settings are displayed in Table 1, and the structure is shown in Fig. 3.

(1) Multi-sensor Fusion Layer (MFL)

Samples from the different m -single sensors are parallelly input to the basic CNN feature extraction layers to obtain the shallow fault information in the training stage, which can be expressed as $z_m = g_m(x_i^m)$, where g_m is the corresponding convolutional layer. To further mine the fault information, the representation of z_m is concatenated with an all-one tensor before the different sensor features are fused. This enables it to model the intra relationship between features in a single vector, denoted as:

$$z'_m = z_m \oplus \mathbf{1}, z_m \in \mathbb{R}^{d_n} \quad (4)$$

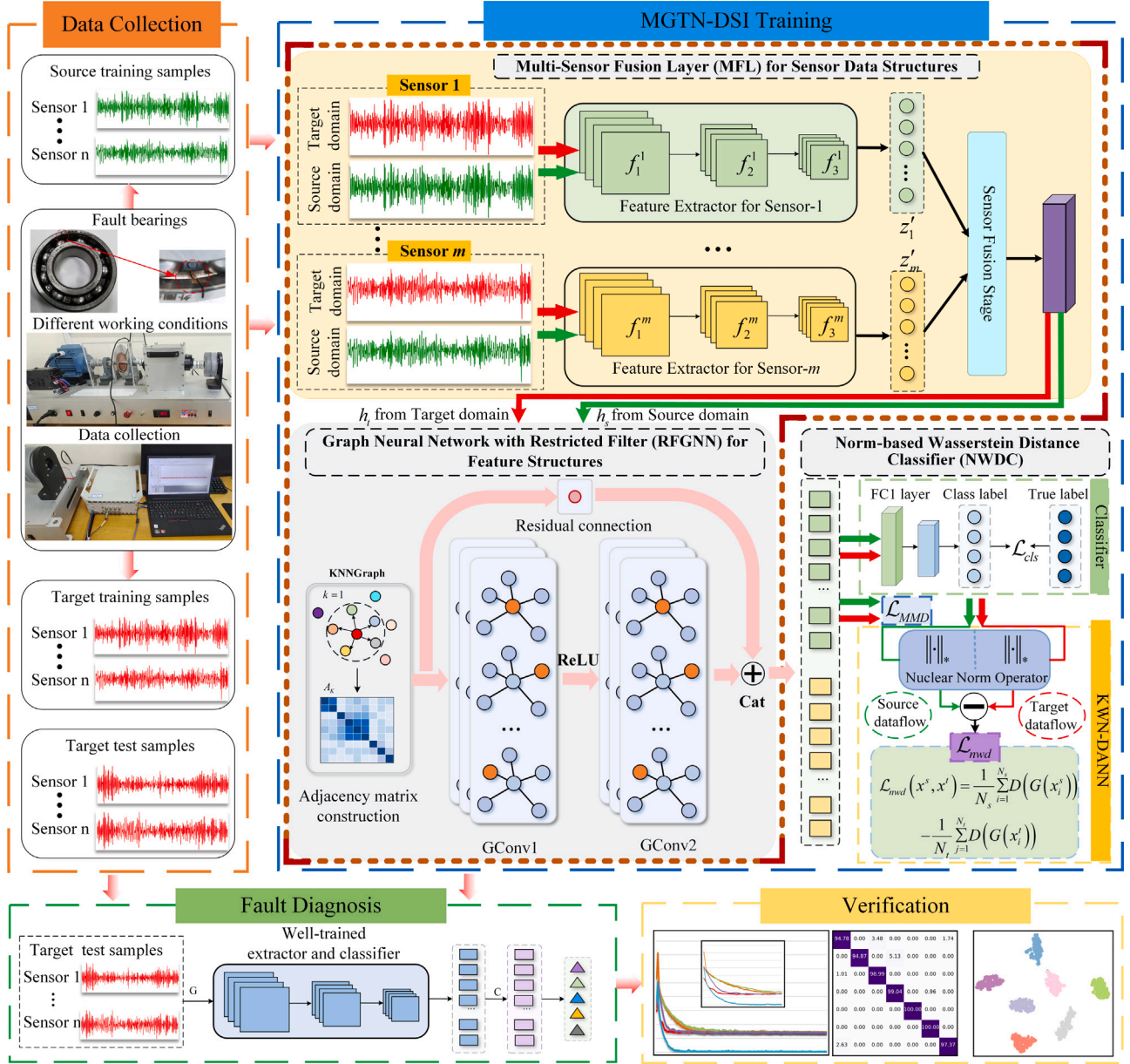


Fig. 2. The Fault diagnosis process structure of the MGTN-DSI Method.

For all signals, different weight parameters W_m are required for the corresponding z'_m . We introduce a trainable weight parameter W to facilitate the multiplication operation with z'_m . Furthermore, to reduce the heavy computational cost due to the increase in the number of sensors, the weight parameters are divided into R parts. Thus, the corresponding weight W_m for m th sensor can be expressed as:

$$W_m = \sum_{i=1}^R w_m^{(i)} \quad (5)$$

where R is the minimum rank that makes the tensor decomposition valid, and $w_m^{(i)}$ represents the i th component of the weight parameter for the m th sensor, derived through the rank R decomposition of W_m . Then z'_m is multiplied with different R learnable weight parameter W_m . Ultimately, the output from the multi-sensor fusion layer is defined by

the following equation:

$$\begin{aligned} h &= W_m \cdot z'_m \\ &= \left(\sum_{i=1}^R w_m^{(i)} \cdot z'_1 \right) \circ \dots \circ \left(\sum_{i=1}^R w_m^{(i)} \cdot z'_m \right) \\ &= \prod_{m=1}^M \left[\sum_{i=1}^R w_m^{(i)} \cdot z'_m \right] \end{aligned} \quad (6)$$

where $\prod_{m=1}^M$ denotes the element-wise production of the tensors.

(2) Graph Neural Network with Restricted Filters (RFGNN)

The fused features obtained by MFL from multi-sensor data are represented as h , where the row vectors in h are regarded as the representations of nodes in the graph. To model the dynamic relationships among features during network training, the Euclidean distance is employed to quantify interactions between nodes, and the distance

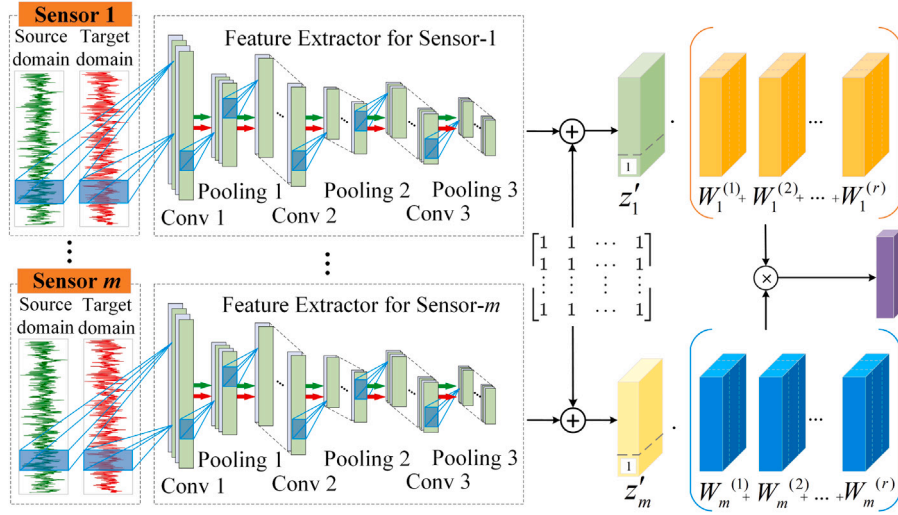


Fig. 3. Multi-sensor features fusion layer.

matrix is expressed as E . Obviously, E is symmetric, with E_{ij} representing the distance between two nodes. To improve the sparsity of the vectors, the smaller k values in each row are designated as 1 via a top- k sorting mechanism, with all other values set to 0. This process involves constructing the adjacency matrix A_K by selecting the K -nearest neighbors for each node, which is calculated by:

$$\begin{cases} E_{ij} = \sqrt{\sum_{n=1}^d (h(i, n) - h(j, n))^2} \\ A_K = \min_k(E) \end{cases} \quad (7)$$

The existing fault diagnosis methods explored the structural relationship between features by utilizing graph neural network architectures. However, the widely used models, such as GNN or ChebNet, tended to learn over-fitting coefficients during training and resulted in compromised diagnostic performance. This susceptibility largely stems from the lack of constraints on the Chebyshev coefficients c_k as indicated in Eq. (3) [34]. To prevent the over-fitting phenomenon and learn accurate coefficients, a novel GNN based on interpolating polynomials has been developed. This approach involves reparameterizing the filter $l(x_j)$ from Eq. (3) into the trainable parameter γ_j . Implementing this methodology enhances the efficiency of the fitting results, allowing for an improved expression of the Chebyshev interpolation polynomial:

$$Q = \frac{2}{K+1} \sum_{k=0}^K \sum_{j=0}^K \gamma_j T_k(h_j) T_k(\hat{L}) h \quad (8)$$

where γ_j represents the learnable filter that is parametrized as above, and $\hat{L}$ is the normalized Laplacian matrix defined as $\hat{L} = I - D^{-1/2} A D^{-1/2}$, where I is the identity matrix and $D_{ii} = \sum_j A_{ij}$ is the degree matrix with diagonal entries representing the degrees of nodes.

To fully aggregate the valid information in the neighborhood, the RFGNN layer is extended to multiple receptive fields to aggregate diverse information from different numbers of neighbors [35], where the structure is shown in Fig. 4.

3.3. Norm-based Wasserstein Distance Classifier (NWDC)

The network structure and optimization goal of the basic DANN are introduced in part 2.2 of Section 2, where the core of the adversarial training is conducting and optimizing a discriminator to enhance its ability to identify the differences between features from two domains. As discussed in [36], the intra-class and inter-class relevance in the prediction outcomes in the classifiers tend to be different for source samples and target samples. Specifically, it is more centralized in the

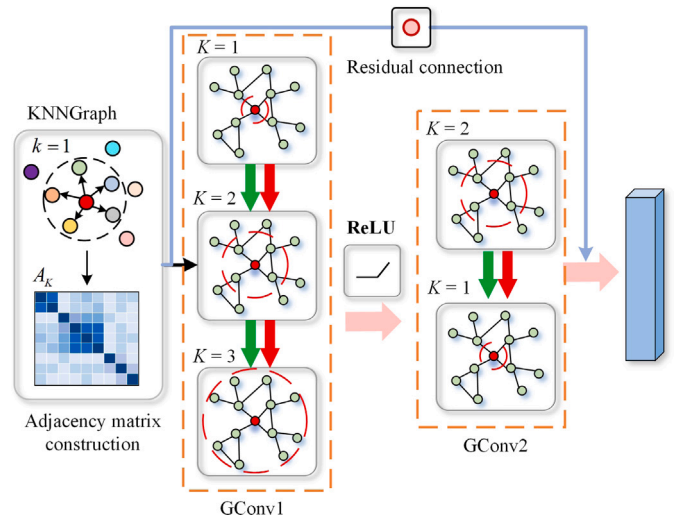


Fig. 4. Detailed structural design of RFGNN.

main diagonal of the self-correlation matrices for the classification outcomes of source samples than target samples. Therefore, the classifier is competent in assessing feature discrepancies between the two domains, which is the goal of the discriminator.

In addition, the Frobenius-norm that is embedded into Wasserstein distance has the ability to measure the discriminability difference of the prediction results from two different domains, meaning that the classifier can also recognize the background domain from which the features were extracted. However, due to the tendency that the Frobenius-norm pushes the model to predict categories with small samples towards the other categories with more samples, the above reused classifier suffers insufficient diversity. Fortunately, maximizing the Nuclear-norm $\|Z\|_*$ can actually increase the rank of matrices and enhance prediction diversity [37]. Finally, the Frobenius-norm is replaced by Nuclear-norm in Wasserstein distance, which can be used in the classifier and thus serve as the discriminator:

$$W_N = \sup_{\|C\|_* \leq K} \mathbb{E}_{\tilde{D}_s} [\|C(f)\|_*] - \mathbb{E}_{\tilde{D}_t} [\|C(f)\|_*] \quad (9)$$

where W_N denotes the Nuclear-norm based Wasserstein (NNW) discrepancy metric for measuring two distributions.

The feature extractor designed extracts features containing complex dual-structure information, which may exacerbate the imbalance

Table 2
Details of fault diagnosis tasks on PU dataset.

Name	Tasks	Work conditions	Dataset	Sensors
PU	A	1500 rpm, 0.7 Nm, 1000 N	H (K004),	Phase Current-1
	B	1500 rpm, 0.1 Nm, 1000 N	OR (KA04, KA30), IR (KI16, KI21),	Phase Current-2
	C	1500 rpm, 0.7 Nm, 400 N	IR+OR (KB23, KB27)	Vibration

between domain alignment and feature discriminability. To enhance cross-domain consistency while ensuring the certainty and diversity of classifier predictions, the NNW is embedded into the classifier C, a virtual discriminator can be established and defined as $D = \|C\|_*$. Thus, the domain identification loss of the virtual discriminator can be calculated from the norm-based Wasserstein distance classifier, expressed as:

$$\mathcal{L}_{NNW}(x^s, x^t) = \frac{1}{N_s} \sum_{i=1}^{N_s} D(G(x_i^s)) - \frac{1}{N_t} \sum_{j=1}^{N_t} D(G(x_j^t)) \quad (10)$$

where $G(\cdot)$ represents the proposed Feature Extractor that considers Dual Structure Information.

The gradient reversal layer that is used in DANN has the ability to update the model in one backpropagation step, which is also adopted in NWDC as shown in Fig. 2.

3.4. Overall objective and optimization

To fully mine the task-specific information under varying working conditions and achieve satisfactory diagnostic results in other tasks, a hybrid loss function that incorporates classification and domain alignment is constructed for the network designed above.

To initialize the model ability to discriminate features, the classification loss of the model is calculated based on the cross entropy, denoted as $\mathcal{L}_c$.

$$\mathcal{L}_c(x^s, y^s) = -\frac{1}{N_s} \sum_{i=1}^{N_s} y_i^s \log(C(G(x_i^s))) \quad (11)$$

At the same time, the classifier is reused as the discriminator with the help of NNW to extract domain-invariant fault features and train the model in an effective but simple way. Moreover, MMD is added to the feature alignment loss for better adaptation. Finally, the overall optimization goal is:

$$\min_{C, G} \left\{ \mathcal{L}_c(x^s, y^s) + \alpha \mathcal{L}_{MMD}(x^s, x^t) + \beta \max_C \mathcal{L}_{NNW}(x^s, x^t) \right\} \quad (12)$$

where α and β are the balanced coefficients.

4. Case study

4.1. Datasets

Case (1) Public Fault Diagnosis Dataset (PU dataset): The experimental configuration for the PU dataset [38] is shown in Fig. 5. This study used data from the PU dataset of accelerated-life damage for experimental validation. Both current and vibration signals are acquired at a sampling frequency of 64 kHz, and a vibration signal dataset and two current signal datasets were eventually obtained. In addition to the health (H) state, the fault categories include outer race (OR), inner race (IR), and their combinations. Moreover, each fault type is categorized into two damage levels. According to the motor torque and load, three different working conditions are established as Table 2 shows, which are set to A, B, and C. Six transfer fault diagnosis tasks were constructed under three different condition tasks. Each category contains 400 samples.

Case (2) Practical Diagnosis Test Platform PT500: A practical fault diagnosis platform, PT500, is utilized to conduct additional experiments, providing further evidence of the proposed method's superiority.

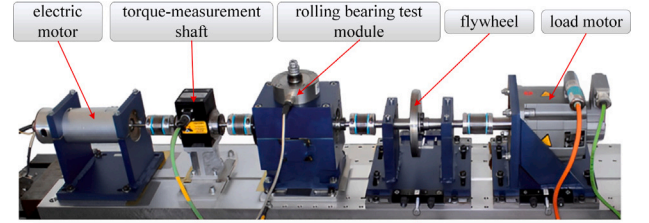


Fig. 5. Acquisition platform for PU dataset.

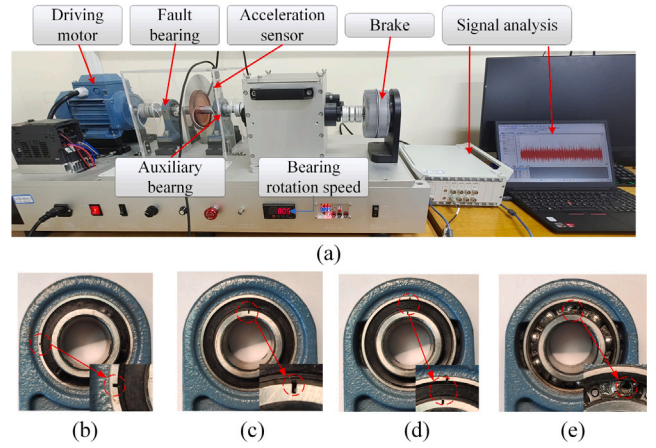


Fig. 6. PT500 bearing fault diagnosis platform and fault types. (a) Experimental platform. (b) Outer race fault. (c) Inner race fault. (d) Compound fault. (e) Ball fault.

Two accelerometers were connected to the fault bearing position and the auxiliary bearing position parallel to it, respectively, and the data was collected at a 10 kHz sampling rate. Five distinct operating conditions are set by varying the drive motor speeds to 800, 1200, 1600, 2000, and 2400 r/min. Fig. 6(b)–(e) shows the four categories of faults except for the normal condition. Each fault category contains 200 samples.

Case (3) BJTU-RAO Dataset [39]: The dataset, which is derived from fault simulation experiments on a 1:2 scale model of a metro train bogie drivetrain conducted by Beijing Jiaotong University, includes fault data from two axle boxes, one gearbox, and one traction motor, all collected under nine different operating conditions. As shown in Fig. 7, the dataset contains four bearing fault states and one normal state from the gearbox, each with 300 samples, collected using two sensors: Channel 12 (near the gearbox input shaft) and Channel 15 (near the gearbox output shaft). This dataset validates the practicality and superiority of the proposed MGTN-DSI method in real-world applications. Detailed experimental settings are shown in Table 3.

4.2. Baseline methods

A variety of transfer fault diagnosis approaches are selected for comparison to highlight the advantages of MGTN-DSI, encompassing six methods based on single sensors and four utilizing multiple sensors.



Fig. 7. BJTU-RAO fault diagnosis platform and fault Types.

Table 3

Details of fault diagnosis tasks on BJTU-RAO dataset.

Name	Work conditions	Health state
BJTU-RAO	Tasks T0: 20 Hz–0 kN	G0: Normal
	Tasks T1: 40 Hz–0 kN	G5: Inner race
	Tasks T2: 60 Hz–0 kN	G6: Outer race
		G7: Rolling element
		G8: Cage

(1) CNN: A well-constructed CNN architecture is employed for both feature extraction and fault identification, where the training is conducted with source samples and the validation is performed with target domain samples. To ensure uniform evaluation across all methods, the same CNN configuration is maintained. The architecture of the CNN used is congruent to that described in the MFL, as shown in Table 1.

(2) JAN [40]: By utilizing the Joint Maximum Mean Discrepancy, JAN achieves conditional alignment by minimizing the joint distribution divergence in multi-domain specific layers, enhancing domain adaptation performance in transfer diagnostic tasks.

(3) DANN [32]: The network of DANN includes an extractor for fault features, a classifier for fault categories, and a domain discriminator that identifies features across domains. With the reversed gradient from the discriminator, the extractor learns to obtain domain-invariant features.

(4) CDANE [41]: Entropy-based Conditional Adversarial Domain Adaptation achieves feature alignment of the same fault category under different domains by an adversarial training procedure using the feature information and label information simultaneously.

(5) DAGCN [27]: Adversarial Domain Adaptive Network (DAGCN) exploits data structure information to enable domain-invariant feature extraction, thereby enhancing the fault diagnosis capabilities across various transfer tasks.

(6) TPC [42]: By minimizing prediction category confusion in the target domain and maximizing the prediction matrix norm, the discriminative ability and diversity can be kept in the TPC model, ensuring the diagnostic performance in transfer tasks.

The following four fault diagnosis methods can learn domain-invariant features under varying working conditions using multi-sensor signals, achieving higher fault identification accuracy.

(7) MFL: To comprehensively evaluate the effectiveness of the multi-sensor information process module, the proposed MLF is used as a baseline method for performance comparison.

(8) DAMFL: By combining MFL with the basic domain adversarial method, i.e., DANN, DAMFL utilizes the multi-sensor information and enhances its domain-invariant feature extraction ability to perform well

in diagnosis tasks under different working conditions.

(9) MDAAN [43]: MDAAN constructs a feature extractor that includes a multi-source information fusion module to facilitate cross-domain FD under the UDA framework.

(10) MSCNN [44]: A multi-scale feature extractor is designed in MSCNN to capture the rich fault features from multi-sensor data, and then address domain bias induced by various working conditions through domain adversarial training.

4.3. Implementation details

During the training phase, the Adam optimizer is utilized. An initial learning rate of 0.0015 is set, which is then reduced by a factor of ten at the 150th and 250th epochs. To minimize the effects of randomness, each method is tested across ten independent trials. All experiments are conducted in the PyTorch environment. As to the Chebyshev polynomial in the proposed MGTN-DSI, three different orders are set to 1, 2, and 3 for multiple receptive fields.

4.4. Results and analysis of PU dataset

(1). Classification Results: Tables 4 and 5 present the results of model comparison experiments conducted using the PU dataset, where a 10-fold cross-validation method is used to minimize experimental errors arising from differences in data splits. Table 4 displays the classification results on the PU dataset utilizing single vibration signals. Across six diverse transfer tasks, an average accuracy of 97.51% has been achieved by MGTN-DSI, which surpasses the DAGCN method, previously considered sub-optimal, by 6.04%. This validates that the proposed RFGNN is capable of more comprehensively considering the structural relationships among samples and more effectively mining domain-invariant information. As shown in Table 5, MGTN-DSI achieved the highest diagnosis accuracy of 99.17% using multi-sensor signals, demonstrating its superiority over single-sensor based methods. MGTN-DSI attained an accuracy of 99.78% on the A→B task, slightly outperforming both MDANN and MSCNN, which highlights its strong adaptability and diagnostic precision for this task. In the A→C task, MGTN-DSI achieved an impressive accuracy of 98.15%, significantly surpassing other methods, especially MDANN, which recorded 93.81%. This exceptional performance demonstrates the robustness of MGTN-DSI, especially in tasks characterized by significant cross-domain differences. The F1-score of MGTN-DSI was consistently higher than or comparable to that of other methods. In the A→C task, it notably outperformed MDANN by over 5%, achieving an F1-score of 0.9103. These results further demonstrate that the proposed multi-sensor feature extractor can capture more representative fault information, maintaining

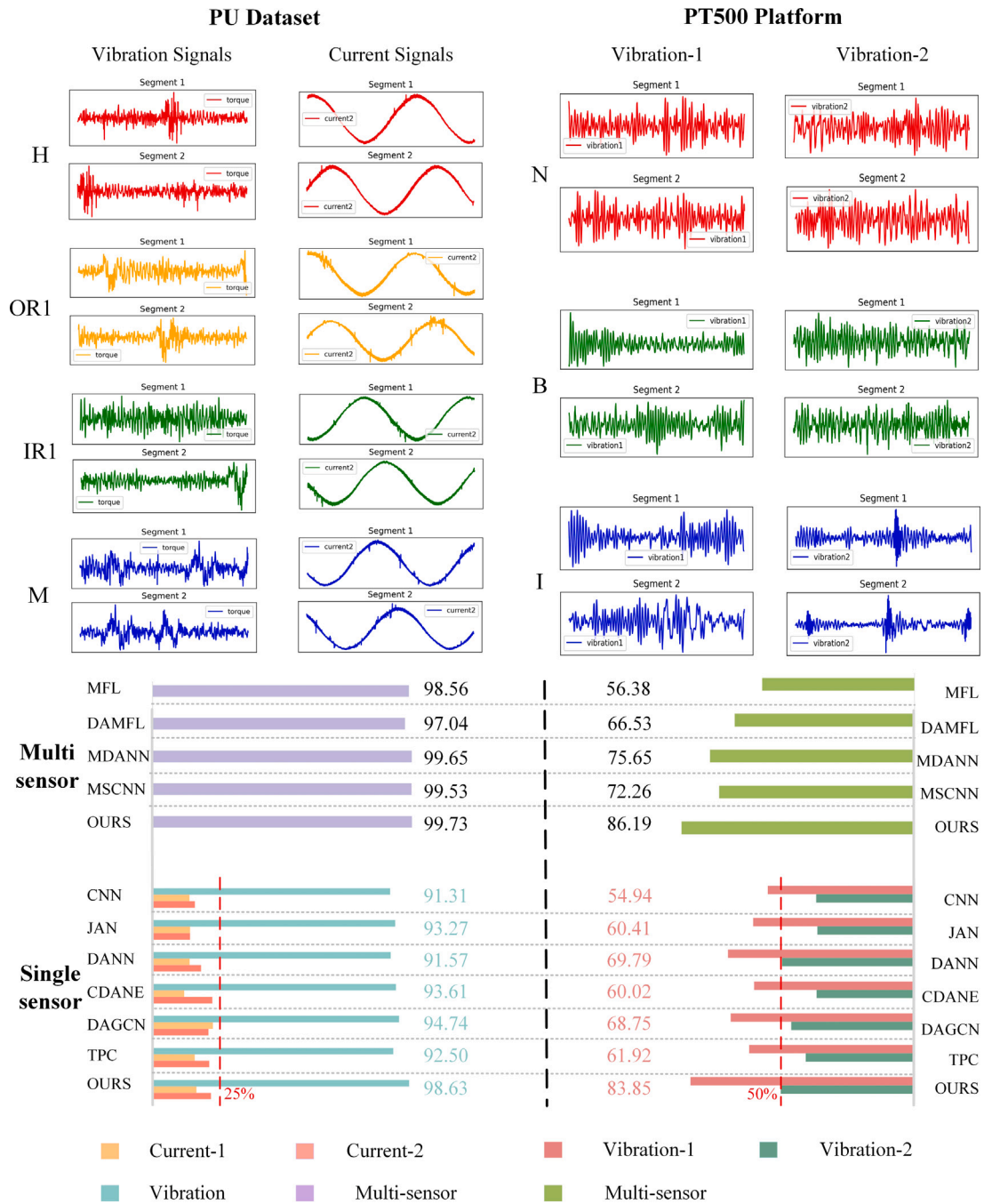


Fig. 8. Visualized different types of signals and the corresponding experiment results. (a) Vibration signals and current signals. (b) Experimental results using different inputs.

Table 4
Classification results on the PU dataset using vibration sensor signals (%).

Task	CNN	JAN	DANN	CDANE	TPC	DAGCN	OURS
A→B	92.48 ± 1.94	94.51 ± 1.83	92.78 ± 1.97	94.67 ± 2.05	93.58 ± 2.65	95.86 ± 1.92	99.88 ± 1.52
A→C	65.94 ± 4.15	79.89 ± 4.25	78.58 ± 4.18	81.88 ± 3.66	78.94 ± 5.21	86.33 ± 2.94	92.94 ± 2.09
B→A	93.92 ± 2.13	95.91 ± 1.95	94.71 ± 2.85	96.18 ± 2.45	94.40 ± 5.43	96.59 ± 1.76	99.78 ± 1.72
B→C	71.96 ± 3.58	81.77 ± 3.02	79.59 ± 4.69	80.92 ± 4.87	82.13 ± 4.86	88.12 ± 2.65	95.29 ± 2.95
C→A	62.36 ± 2.29	87.92 ± 3.19	72.96 ± 4.62	75.18 ± 3.42	81.14 ± 3.64	91.61 ± 4.16	98.18 ± 2.52
C→B	61.55 ± 3.97	86.52 ± 4.65	79.12 ± 3.57	83.22 ± 2.85	78.62 ± 3.83	90.33 ± 2.74	98.97 ± 1.87
Average	74.70 ± 3.01	87.75 ± 3.15	82.96 ± 3.65	85.34 ± 3.22	84.77 ± 4.27	91.47 ± 2.36	97.51 ± 2.11

excellent classification performance even under significantly varying conditions. Additionally, MGTN-DSI exhibits strong feature alignment capabilities, ensuring high-quality diagnostic performance in scenarios involving both single and multiple sensors.

(2). Effectiveness of Different Signal Types: The left part of Fig. 8 visualizes the current signal and the vibration signal used in the experiments on the PU dataset. Compared with the vibration signal, the difference in current signals between variant fault categories is small.

Table 5
Classification results on the PU dataset using multi-sensor signals.

Task	MFL		DAMFL		MDANN		MSCNN		OURS	
	ACC (%)	F1-score	ACC (%)	F1-score	ACC (%)	F1-score	ACC (%)	F1-score	ACC (%)	F1-score
A→B	99.79	0.9102	97.80	0.9308	99.67	0.9348	99.60	0.9510	99.78	0.9485
A→C	71.66	0.8040	90.38	0.8613	93.81	0.8520	87.03	0.8675	98.15	0.9103
B→A	72.16	0.7790	81.57	0.7782	99.75	0.9570	99.63	0.9530	99.70	0.9602
B→C	76.36	0.8392	70.50	0.7260	98.40	0.9025	94.00	0.8895	99.20	0.9025
C→A	72.54	0.8170	96.00	0.9010	97.20	0.9070	95.85	0.9102	98.92	0.9355
C→B	79.44	0.8601	94.40	0.8618	98.20	0.9212	97.98	0.9105	99.25	0.9380
Average	78.95	0.8356	88.53	0.8432	97.84	0.9124	95.68	0.9203	99.17	0.9425

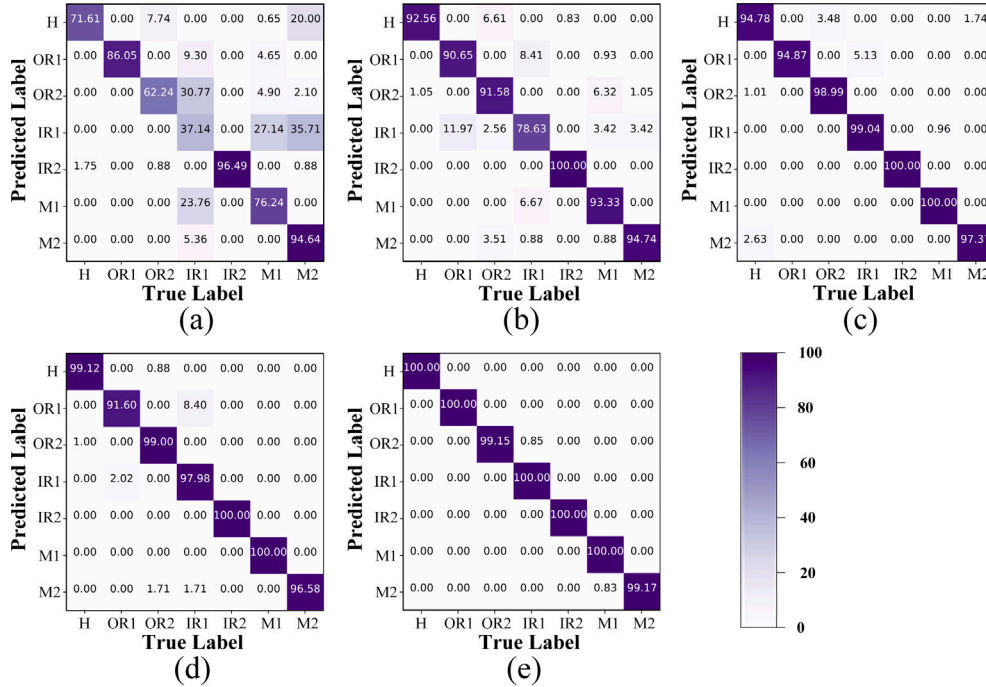


Fig. 9. Confusion matrix for task C → B. (a) MFL. (b) DAMFL. (c) MSCNN. (d) MDANN. (e) MGTN-DSI(OURS).

From the experimental results, the vibration signal based methods have better diagnosis results than the current signals based methods, which can be reflected by the diagnostic accuracy of over 90% and under 20% for the two data form-based methods, respectively. Although the discriminability of current signals is almost absent, MFL can further enhance the diagnosis accuracy from 98.63% to 99.73% by utilizing the current signals in addition to the vibration signals, indicating the effectiveness of the designed information fusion layer. Similarly, the vibration signals collected from different locations on the PT500 platform are visualized and the results are shown in the right part of Fig. 8. Compared to the sensor signals of the PU dataset, the two vibration signals of the PT500 platform are more distinguishable. According to the test results, the MFL strategy enhances the overall presentation of the data by multi-sensor fusion, which further validates the advantage of MFL.

(3) Confusion Matrices Analysis: Fig. 9 shows the confusion matrices of our method MGTN-DSI and the multi-sensor comparison methods on task C→B. It should be noted that faults with different damage levels are distinguished by numbers, such as IR1 and IR2, and M represents the combination faults of IR and OR for the convenience of presentation. Obviously, MGTN-DSI perfectly identifies all samples in five fault categories, including H, OR1, IR1, IR2, and M1. Moreover, the classification accuracy on the other two fault categories of the proposed method also exceeds 99%, demonstrating its huge advantages over comparison methods. The above results validate the superiority of constructing feature information and the fine-grained feature alignment that is achieved by the NNW metric.

(4). Feature Visualization: To further analyze different methods, task C→B is chosen at random to display the feature visualization results under the multi-sensor input scenario. As Fig. 10 shows, MGTN-DSI achieves the best domain adaptation effect among all methods, which can be reflected by the fine-grained alignment of features from two domains and the clear separation of different fault categories. Among the comparison methods, MSCNN and MDANN also show competitive results. However, there are still relatively noticeable overlaps in the red boxes of Fig. 10(c)–(d), which is the reason for misclassification. Moreover, DAMFL has better feature alignment performance than the basic MFL, meaning that the adversarial training using the domain discriminator helps to achieve domain adaptation. But MGTN-DSI can further improve alignment performance than DAMFL, which uses the NNW metric to construct a virtual discriminator and train the model, demonstrating the superiority of the norm-based Wasserstein distance classifier.

4.5. Results and analysis on PT500 test bench

(1). Classification Results: Table 6 presents the classification results with single-sensor data, while Table 7 showcases those with multi-sensor data, both obtained from the practical fault diagnosis test platform PT500. Using the single-vibration signals without MFL, the proposed MGTN-DSI achieves the highest average classification accuracy on both two different single-vibration signals and also is the only method that has an accuracy exceeding 90% using Vibration-1, demonstrates the superiority of the feature structure information

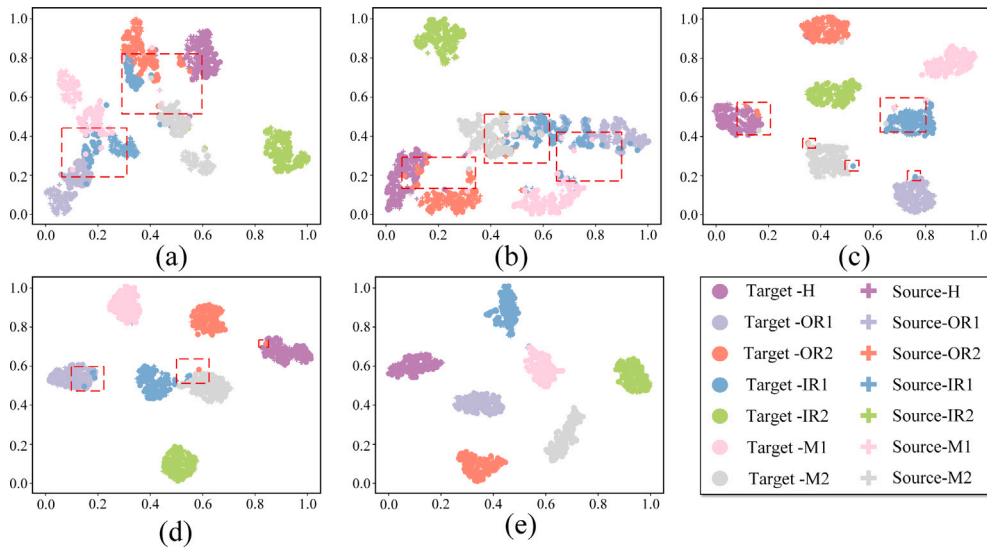


Fig. 10. Feature visualization results of task C → B. (a) MFL. (b) DAMFL. (c) MSCNN. (d) MDANN. (e) MGTN-DSI(OURS).

Table 6

Accuracy of single sensor signals classification for on the PT-500 dataset (%).

Task	Vibration-1							Vibration-2						
	CNN	JAN	DANN	CDANE	TPC	DAGCN	OURS	CNN	JAN	DANN	CDANE	TPC	DAGCN	OURS
L0→L1	83.85	82.81	89.45	83.46	87.60	87.36	97.13	49.34	47.13	55.07	46.74	55.99	58.20	56.55
L0→L2	85.93	86.45	95.83	87.76	95.44	88.80	98.30	50.91	45.83	48.82	42.57	60.33	47.26	56.77
L0→L3	80.85	83.98	93.61	83.98	94.36	85.41	98.04	40.10	39.84	43.75	38.41	52.60	46.35	53.43
L0→L4	70.96	77.21	89.97	79.68	92.84	79.68	94.01	33.85	36.84	37.23	36.71	39.60	38.02	39.65
L1→L0	70.83	71.74	86.71	69.66	68.20	77.47	93.48	42.83	42.44	51.82	41.27	49.66	53.77	46.22
L1→L2	90.36	91.53	97.00	91.14	96.00	95.18	97.65	55.72	57.55	76.95	58.33	75.20	60.28	71.48
L1→L3	85.67	90.36	96.74	89.19	96.32	94.92	96.09	37.50	50.00	58.46	50.26	58.60	55.59	54.81
L1→L4	80.85	84.89	97.39	88.54	95.76	91.53	97.65	42.70	47.91	54.29	45.96	57.00	53.25	49.73
L2→L0	54.94	60.41	69.79	60.02	61.92	68.75	83.85	36.84	36.45	49.60	36.71	40.80	46.22	50.13
L2→L1	79.16	81.11	85.41	79.42	82.76	85.54	90.10	54.29	56.51	68.22	56.96	60.73	61.71	68.75
L2→L3	95.44	95.18	98.56	95.83	98.92	99.21	99.08	65.49	68.75	80.72	68.35	76.20	80.98	86.19
L2→L4	87.10	90.75	99.60	95.31	98.36	98.17	99.60	57.03	61.58	72.65	65.10	67.33	75.00	80.46
L3→L0	48.04	61.58	72.52	58.59	55.11	69.14	69.79	34.37	39.19	40.62	37.63	42.73	43.61	50.26
L3→L1	71.61	76.56	82.68	76.30	73.12	81.90	85.41	40.62	47.39	51.69	47.78	48.60	52.21	61.32
L3→L2	93.22	93.35	96.48	94.79	96.35	97.52	96.35	62.50	64.97	73.95	68.22	67.20	74.47	79.16
L3→L4	97.13	98.17	99.60	93.35	99.64	99.73	99.73	69.66	69.92	80.85	71.87	80.60	80.59	91.92
L4→L0	43.75	60.54	63.54	57.68	48.68	65.23	67.05	26.82	35.02	37.89	36.58	31.73	34.89	30.72
L4→L1	63.15	73.69	81.51	70.05	67.12	79.29	81.77	30.85	45.70	49.08	45.57	45.99	52.73	52.82
L4→L2	82.42	88.02	94.53	83.72	87.24	91.66	94.79	47.00	50.39	60.02	53.64	55.00	61.06	64.84
L4→L3	96.61	96.87	98.56	95.18	98.40	98.82	98.56	62.63	63.02	69.40	63.93	70.53	70.44	76.69
Average	78.09	82.26	89.47	81.68	84.70	86.77	91.92	47.05	50.32	58.05	50.63	56.82	57.33	61.09

extracted by RFGNN, alongside the effective NNW-based alignment. In the multi-sensor experiments, MGTN-DSI achieved the highest average accuracy of 93.69%, surpassing other methods in fifteen out of twenty tasks, showing a consistent lead. Specifically, under tasks with significant variations in working conditions, MGTN-DSI demonstrated a remarkable advantage, outperforming other methods by a margin of 5% to 15% in accuracy. Furthermore, the F1-scores for MGTN-DSI are notably higher, with an average of 0.8967 compared with 0.8693 for MDANN and 0.8650 for MSCNN. In several tasks, such as L0→L1 and L1→L2, the F1-scores of MGTN-DSI exceed 0.9, which indicates its superior capability to maintain a balance between precision and recall. For instance, in the L0→L1 task, MGTN-DSI achieves an F1-score of 0.8987, outperforming MDANN by nearly 0.1, underscoring its ability to accurately identify and classify faults even under complex scenarios. Such balanced performance suggests that the MFL module effectively fuses features, resulting in robust diagnostic capabilities even under challenging conditions. As multi-sensor technologies advance, the ability to gather richer data sets enhances diagnostic performance, positioning MGTN-DSI as a reliable and effective method for practical applications across diverse scenarios, whether using single-sensor or

multi-sensor inputs.

(2). Confusion Matrices Analysis: The confusion matrix results for MGTN-DSI and various multi-sensor comparison methods in task L1→L0 are depicted in Fig. 11. Among all fault diagnosis methods, MDANN, MSCNN, and MGTN-DSI correctly identify all samples with normal state, demonstrating great advantages over other methods. However, MDAAN and MSCNN misclassified more than 10% and 18% samples of the ball fault, respectively. In contrast, the classification accuracy in the ball fault of MGTN-DSI reaches 96%. Moreover, the diagnosis accuracy of MGTN-DSI in each fault category is no less than 92%, illustrating the effectiveness of the proposed method.

(3) Feature Visualization: Fig. 12(a) visualizes the fault features extracted by the baseline method MFL, where the target features in the red box overlap more seriously compared to the other four methods, leading to poor diagnosis performance. For comparison, the other three comparison methods can effectively reduce this phenomenon, but there is still a certain degree of overlapping areas in different category features in Fig. 12(b)–(d). By extracting dual structure information and aligning features with a virtual discriminator, MGTN-DSI successfully aligns fault features of different categories from two domains, achieving

Table 7

Accuracy of multi-sensor signals classification on the PT-500 dataset.

Task	MFL		DAMFL		MDANN		MSCNN		OURS	
	ACC (%)	F1-score	ACC (%)	F1-score	ACC (%)	F1-score	ACC (%)	F1-score	ACC (%)	F1-score
L0→L1	86.97	0.7738	86.97	0.7857	90.88	0.7991	91.40	0.8205	95.83	0.8987
L0→L2	89.32	0.7479	87.63	0.8089	97.34	0.8570	93.75	0.8452	97.52	0.9101
L0→L3	94.76	0.7756	86.06	0.7710	96.32	0.8510	95.57	0.8722	96.74	0.8959
L0→L4	83.46	0.7040	79.03	0.7860	96.09	0.8346	93.35	0.8664	97.39	0.8879
L1→L0	73.43	0.7560	75.78	0.7657	83.85	0.7928	81.51	0.7978	95.31	0.8834
L1→L2	92.96	0.8491	93.48	0.8547	98.69	0.8767	95.31	0.8789	97.26	0.9226
L1→L3	92.18	0.8122	92.44	0.8610	97.42	0.8670	95.44	0.8891	98.04	0.9063
L1→L4	86.19	0.7972	87.63	0.8060	97.00	0.8837	94.53	0.8823	98.56	0.9182
L2→L0	56.38	0.7054	66.53	0.7106	75.65	0.8229	72.26	0.7408	86.19	0.8090
L2→L1	81.90	0.8405	82.94	0.8066	84.24	0.8504	85.80	0.8747	91.66	0.9231
L2→L3	93.88	0.8987	94.79	0.8889	99.86	0.9435	98.69	0.9273	99.73	0.9446
L2→L4	91.01	0.8849	94.79	0.8687	99.60	0.9333	99.08	0.9304	99.60	0.9384
L3→L0	54.03	0.7068	66.27	0.7378	74.34	0.8155	68.88	0.7889	82.16	0.8025
L3→L1	72.78	0.7802	76.95	0.8012	79.16	0.8474	80.46	0.8542	87.76	0.8820
L3→L2	94.01	0.9177	94.14	0.8886	98.95	0.9370	96.09	0.9102	98.17	0.9213
L3→L4	97.26	0.9235	95.96	0.9210	100.00	0.9455	99.86	0.9396	100.00	0.9460
L4→L0	47.91	0.6704	64.32	0.7765	69.53	0.7963	68.35	0.7847	72.64	0.7876
L4→L1	71.48	0.7753	75.52	0.8032	78.90	0.8456	79.29	0.8592	84.24	0.8681
L4→L2	84.24	0.8448	90.36	0.8990	98.17	0.9384	93.22	0.9030	96.09	0.9351
L4→L3	94.14	0.9067	94.66	0.9068	99.86	0.9490	99.08	0.9336	99.06	0.9523
Average	81.41	0.8035	84.31	0.8224	90.80	0.8693	89.10	0.8650	93.69	0.8967

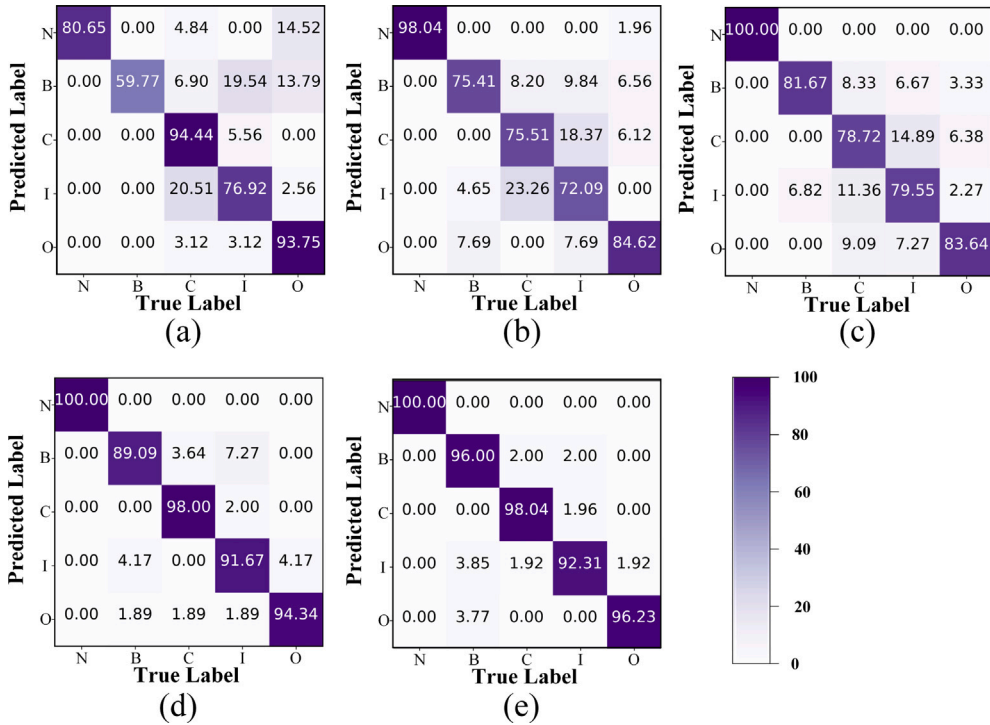


Fig. 11. Confusion matrix for task L1 → L0. (a) MFL. (b) DAMFL. (c) MSCNN. (d) MDANN. (e) MGTD-DSI(OURS).

satisfactory fault diagnosis results.

(4) Ablation Study: The performance of various modules in MGTD-DSI is verified and comprehensive ablation experiments are conducted on the PT500 fault diagnosis test platform. Four ablation comparison methods are designed by removing or replacing different modules, with the diagnosis results presented in Table 8. First, the baseline method is constructed by obtaining feature structure information with CNN and ChebNet and achieving domain adaptation by adversarial training, whose diagnosis accuracy is 86.34% on the practical platform PT500. Using the designed MFL module, Method A increases the diagnostic accuracy by 2.76%, manifesting the importance of the information fusion layer. In addition, Method B is established by replacing the basic

ChebNet module with the RFGNN module. The corresponding fault diagnosis result is 2.09% higher than Method A, meaning that it is necessary to reparametrize the filter with interpolating polynomials and extract dual structure information. Finally, Method C, i.e., MGTD-DSI utilizes the NNW metric based virtual discriminator to achieve feature alignment, whose accuracy is 93.68% and exceeds Method B with a margin of 2.49%. Therefore, the reused classifier with the NNW metric is more suitable for aligning features with dual structures compared to traditional domain adversarial training procedures. To summarize, each module is necessary for MGTD-DSI to achieve satisfactory performance in transfer fault diagnosis tasks.

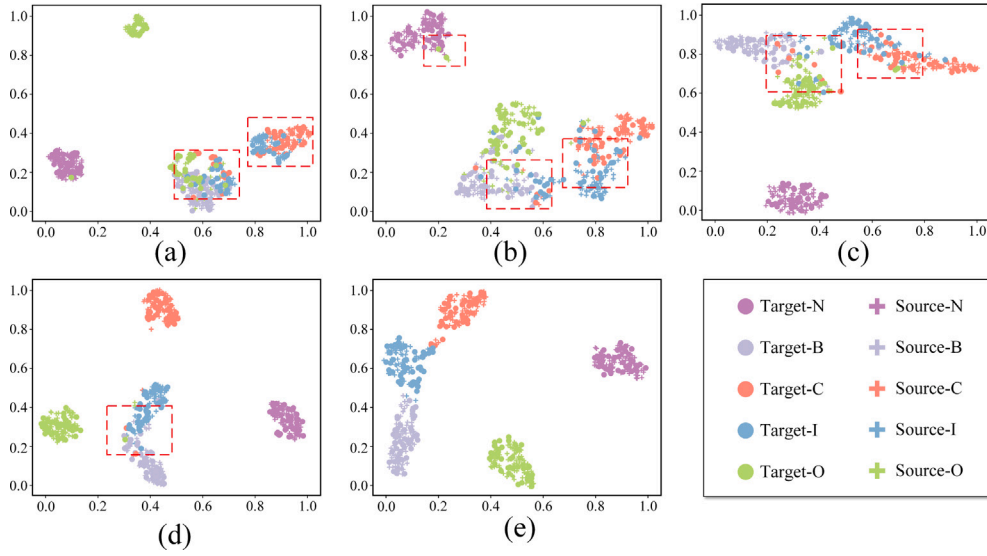


Fig. 12. Feature visualization results of task L1 → L0. (a) MFL. (b) DAMFL. (c) MSCNN. (d) MDANN. (e) MGTN-DSI(OURS).

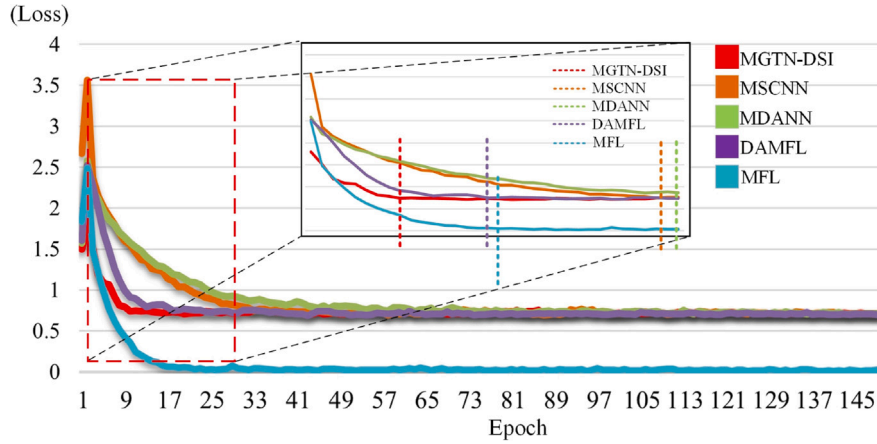


Fig. 13. Loss curves of different methods on task L3→L0.

Table 8

Ablation studies of the proposed method.

Modules	Baseline	Method A	Method B	Method C
CNN	✓			
ChebNet	✓	✓		
DANN	✓	✓	✓	
MFL		✓	✓	✓
RFGNN			✓	✓
NWDC				✓
Average (%)	86.34 (-)	89.10 (2.76↑)	91.19 (4.85↑)	93.68 (7.34↑)

(5) Convergence Speed Comparison: To further validate the superiority of MGTN-DSI, the loss functions for five multi-sensor approaches in the L3→L0 task are displayed in Fig. 13. It should be noted that the final loss of MFL is lower than the other methods because the optimization goal of MFL only contains the classification loss. Obviously, MGTN-DSI demonstrates more rapid convergence compared to four benchmark methods, underscoring the enhanced optimization capability of this approach. Such optimization efficiency substantiates its viability for deployment in real-world fault diagnosis applications.

4.6. Results and analysis of BJTU-RAO dataset

(1). Classification Results: To validate the effectiveness of our proposed MGTN-DSI method, we utilized the BJTU-RAO dataset, which reflects real-world application scenarios, and conducted experiments using 10-fold cross-validation. Table 9 shows the classification results of various multi-sensor methods. MGTN-DSI consistently achieves the highest performance across all transfer tasks, demonstrating robustness and effectiveness in handling multi-sensor signals under varying conditions. For example, in task T0→T1, the proposed method achieves an accuracy of 98.90% and an F1-score of 0.9648, outperforming the next best method, MSCNN, which reaches 97.42% accuracy and an F1-score of 0.9379. In challenging tasks like T2→T0, where baseline methods struggle, our method excels with 92.26% accuracy compared to MFL's 51.79%, highlighting its superior feature alignment capabilities. Furthermore, the proposed method demonstrates superior consistency, achieving the highest average accuracy (97.51%) and F1-score (0.9482) across all tasks. Even in competitive scenarios, such as T1→T2, our approach maintains its lead with 99.92% accuracy and an F1-score of 0.9593, surpassing MDANN's 99.14% accuracy. These results underscore the effectiveness, stability, and practical applicability of the proposed method for multi-sensor fault diagnosis under varying working conditions.

(2) Visual analysis of results: Fig. 14 presents the comparison results between various multi-sensor methods and our proposed MGTN-DSI.

Table 9
Classification results on the BJTU-RAO dataset.

Task	MFL		DAMFL		MDANN		MSCNN		OURS	
	ACC (%)	F1-score	ACC (%)	F1-score	ACC (%)	F1-score	ACC (%)	F1-score	ACC (%)	F1-score
T0→T1	79.53	0.8683	95.39	0.9059	96.79	0.9258	97.42	0.9379	98.90	0.9648
T0→T2	74.60	0.8277	90.23	0.9053	93.75	0.9379	86.32	0.7508	98.04	0.9477
T1→T0	74.60	0.8090	80.70	0.8672	94.06	0.9117	95.23	0.9199	97.50	0.9468
T1→T2	92.34	0.9148	95.15	0.9000	99.14	0.9357	98.90	0.9411	99.92	0.9593
T2→T0	51.79	0.7291	79.60	0.8417	86.04	0.8507	81.32	0.8629	92.26	0.9118
T2→T1	74.53	0.8802	85.78	0.9166	91.47	0.9245	97.10	0.9298	98.43	0.9585
Average	74.57	0.8385	87.81	0.8894	93.54	0.9144	92.72	0.8904	97.51	0.9482

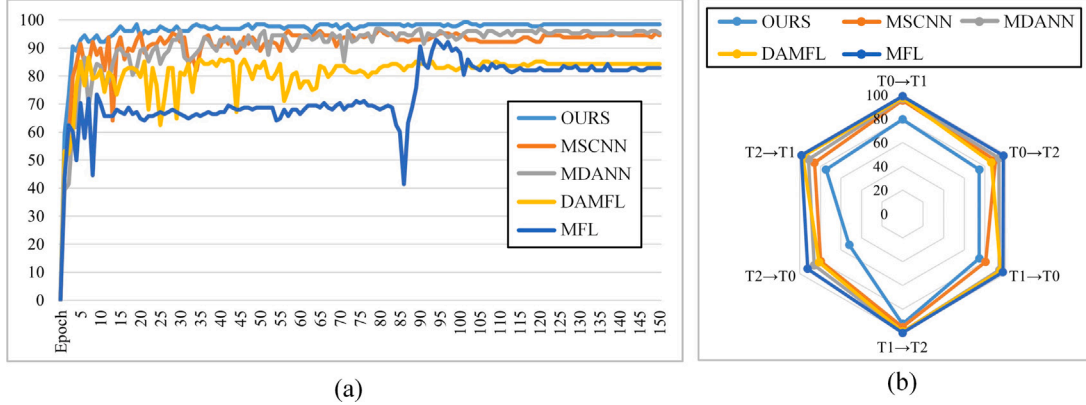


Fig. 14. Multi-sensor comparison experiment results under task T1→T0 for the BJTU-RAO dataset (a) Model training visualization (b) Radar plot of multi-sensor experiment results.

It is worth noting that the result displayed in Fig. 14(a) is from a randomly selected trial of the T1→T0 task. Fig. 14(a) illustrates the training processes of different methods. The proposed method not only maintains the highest fault identification accuracy but also demonstrates faster convergence compared to the second-best methods, MSCNN and MDANN. This result further proves its superiority and strong performance in cross-condition practical applications. Additionally, Fig. 14(b) presents the detailed performance metrics of five models under various operating conditions. For the T0→T1 task, all methods achieve relatively high accuracy. However, as the degree of operational condition variation increases, the performance of all methods declines to some extent. Notably, our method exhibits the smallest performance degradation. In particular, for the T2→T0 tasks, the baseline method MFL struggles to complete the fault recognition task, while the recognition accuracy of the second-best methods, MSCNN and MDANN, shows a more notable drop compared to our method. This underscores the effectiveness of our dual-structure information-based feature extractor, which adeptly integrates data from multiple sensors and extracts deeper domain-invariant features. Combined with our joint domain alignment strategy, this approach significantly enhances fault recognition precision and reliability in real-world scenarios.

In summary, the proposed MGTN-DSI method also performs exceptionally well on the BJTU-RAO experimental platform, which simulates real-world application scenarios. These results validate the generalization ability and reliability of the proposed method.

4.7. Model parameter and complexity analysis

(1). Rank Selection: The hyperparameter R is the minimum rank that makes the decomposition valid in the designed MFL module, influencing the feature representation capability and the computational cost in the multi-sensor data fusion process. To determine an appropriate value for R , we performed relevant experiments using the PU dataset. As shown in Fig. 15, there is a valley in the curve of standard deviation, i.e., the model is most stable when R is 4. In addition, the diagnosis accuracy is almost near 86% under different R except 16 and 32. Therefore, R is taken as 4 in this study to achieve satisfactory fault

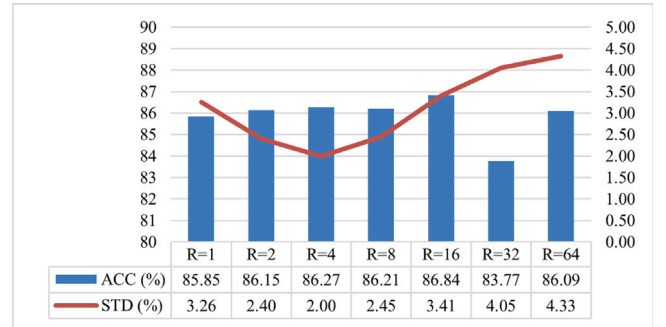


Fig. 15. Experimental results under different rank choices.

diagnosis accuracy with a small standard deviation.

(2). Layer Design of RFGNNs: The layer design experiment aims to determine the optimal number of layers for the RFGNN model. The experiments are carried out on the practical diagnosis test platform PT500, where a transfer task (L1→L2) has been randomly selected as an example. We increased the number of layers n in GConv2 from 1 to 6. To maintain fairness and comparability, each set of experiments is repeated 5 times, with all other parameters remaining consistent with those specified in the preceding text. As shown in Fig. 16, the performance of the model initially improves with an increase in the number of layers but then declines. When the number of layers is 2 or 3, the model achieves relatively stable results. However, increasing the number of layers further may lead to overfitting, which negatively impacts diagnostic accuracy. Since the performance is almost identical when using 2 or 3 layers, we have selected to set Gconv2 to 2 layers to reduce the computational cost.

(3). Complexity Analysis Experiment: Table 10 presents the training and inference times for state-of-the-art methods on the PU dataset. Our proposed method has an average training time of 4.0646 s, primarily due to the additional computations required for constructing graph structures in its GNN-based feature extraction process. Despite this, it

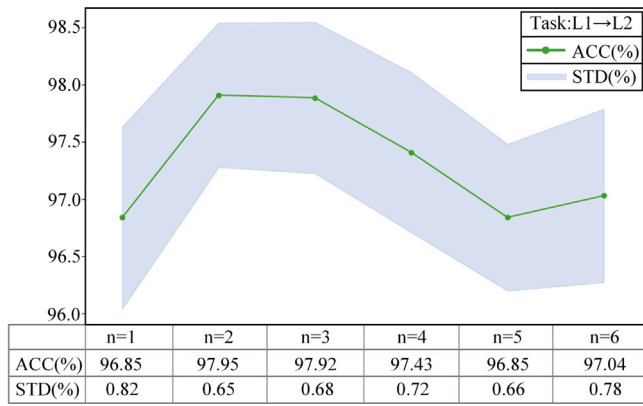


Fig. 16. Experimental results under different layer choices.

Table 10
Training time and inference time for PU dataset.

Method	Average training time (s)	Average inference time (s)
OURS	4.0646	0.0231
DAGCN	3.3095	0.0200
TPC	0.5900	0.0010
CDANE	1.1943	0.0037
DANN	1.1029	0.0035
JAN	1.2552	0.0036
CNN	1.0016	0.0037

achieves a competitive inference time of approximately 0.023 s. Our method demonstrates a significant improvement in diagnostic accuracy, achieving approximately a 10% enhancement over most baseline methods. This performance is attributed to two key factors: (1) Effective Utilization of Structural Information: Integrating multi-sensor data allows for richer and more diverse feature representations; (2) Optimized Domain Adaptation: A tailored domain adaptation mechanism enhances generalization across varying operational conditions.

In summary, while our method requires slightly longer training times, it excels in diagnostic accuracy and maintains competitive inference times. These features highlight its robust feature extraction capabilities and overall effectiveness. Thus, our approach offers a superior solution for fault diagnosis tasks, demonstrating substantial advantages and reliability in practical applications.

5. Conclusion

In this research, we present a novel multi-sensor graph transfer learning network, MGTN-DSI, to address two challenges in cross-domain fault diagnosis tasks, including the difficulty of efficiently exploiting structural information in multi-sensor data and the issue of imbalance between domain alignment and feature discriminability. Specifically, we design a feature extractor that considers dual structural information to obtain domain-invariant features from two perspectives. Additionally, an adversarial method that utilizes a reused classifier with a Nuclear-norm based Wasserstein metric is designed, and by integrating a metric-based domain discrepancy loss, a new joint domain distribution alignment method is developed, thus facilitating the discernibility between distinct domains while markedly improving the capacity for feature alignment. The effectiveness of the proposed method was validated through a series of fault diagnosis experiments. The results showed that MGTN-DSI achieved better performance in comparative experiments using multi-sensor data from both the PU dataset and the practical fault diagnosis test platform PT500, with accuracies of 99.17% and 93.69%, respectively. Additionally, compared to the state-of-the-art comparison models, MGTN-DSI obtained the

highest F1-scores in both experiments, reaching 0.9425 and 0.8967. These results demonstrate the superior fault diagnosis capabilities of the proposed method, proving its ability to capture and align features across diverse sensors, leading to outstanding diagnosis performance under varying conditions.

Future investigations might focus on devising hypergraphs for multi-sensor data and refining MGTN-DSI to integrate hypergraph data more effectively.

CRedit authorship contribution statement

Jianjie Liu: Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Xianfeng Yuan:** Writing – review & editing, Supervision, Resources, Funding acquisition, Conceptualization. **Xilin Yang:** Writing – review & editing, Visualization. **Weijie Zhu:** Writing – original draft, Data curation. **Yansong Zhang:** Visualization, Investigation. **Tianyi Ye:** Validation, Supervision. **Xinxin Yao:** Visualization, Supervision. **Fengyu Zhou:** Supervision, Project administration, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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